

Computer Networks

What is the OSI Model?

The OSI (Open Systems Interconnection) model is a conceptual framework used to understand network interactions in seven layers. Each layer serves the layer above it and is served by the layer below it. The seven layers, from bottom to top, are: Physical, Data Link, Network, Transport, Session, Presentation, and Application. The model helps standardize communication functions so that different systems and vendors can interoperate over a network.

The Seven Layers Explained

The Physical layer deals with the actual transmission of raw bits over a physical medium such as cables or radio waves. The Data Link layer handles node to node data transfer and error detection using MAC addresses. The Network layer is responsible for routing packets across different networks using logical addressing, primarily IP addresses. The Transport layer ensures reliable data transfer between hosts using protocols like TCP and UDP, and manages flow control and error recovery. The Session layer manages sessions between applications, establishing, maintaining, and terminating connections. The Presentation layer translates data between the application and network formats, handling encryption and compression. Finally, the Application layer is closest to the end user and provides network services directly to applications such as web browsers and email clients.

TCP vs UDP

TCP (Transmission Control Protocol) is connection oriented, meaning it establishes a connection before transferring data using a three way handshake. It guarantees reliable, ordered delivery of data and includes error checking and retransmission of lost packets, making it suitable for applications like file transfer and web browsing where accuracy matters more than speed. UDP (User Datagram Protocol) is connectionless and does not guarantee delivery, ordering, or error checking. It is faster and has lower overhead than TCP, making it suitable for applications like video streaming, online gaming, and DNS lookups where speed matters more than perfect reliability.

IP Addressing

An IP address is a unique numerical identifier assigned to every device on a network. IPv4 addresses are 32 bits long, typically written as four decimal numbers separated by dots, such as 192.168.1.1, giving approximately 4.3 billion unique addresses. Due to the exhaustion of IPv4 addresses, IPv6 was introduced, using 128 bit addresses written in hexadecimal notation, which provides a vastly larger address space. IP addresses can be classified as public, which are routable on the internet, or private, which are used within local networks and are not directly reachable from the internet.

Subnetting

Subnetting is the process of dividing a large network into smaller sub networks, or subnets. This is done using a subnet mask, which determines how many bits of an IP address represent the network portion versus the host portion. Subnetting improves network performance by reducing broadcast traffic, and improves security by isolating segments of a network from each other. For example, a Class C network with the default mask 255.255.255.0 can be subnetted further into smaller ranges depending on organizational needs.

DNS (Domain Name System)

DNS translates human readable domain names, such as example.com, into IP addresses that computers use to identify each other on a network. Without DNS, users would need to remember numerical IP addresses for every website they wanted to visit. The DNS lookup process typically involves a recursive resolver, root name servers, top level domain servers, and authoritative name servers, each providing part of the information needed to resolve a domain name to its corresponding IP address. DNS also uses caching extensively to reduce lookup times for frequently visited domains.

Firewalls

A firewall is a network security device or software that monitors and controls incoming and outgoing network traffic based on predetermined security rules. Firewalls establish a barrier between a trusted internal network and untrusted external networks, such as the internet. They can operate at different layers of the OSI model, with packet filtering firewalls working at the network layer and application layer firewalls inspecting the actual content of traffic for specific applications.